

122. The tendency of 3d-metal ions to form stable complexes is due to their

- (a) Variable oxidation state
- (b) Strong electronegative nature
- (c) High charge/size ratio and vacant d-orbitals**
- (d) Very low ionization energies

Answer: (c) High charge/size ratio and vacant d-orbitals

Explanation: Transition-metal ions form complexes readily because of their relatively high charge density and available orbitals for accepting ligand electron pairs.

123. The 3d-metal ions are paramagnetic in nature because

- (a) They are reducing agents
- (b) They form coloured salts
- (c) They have one or more paired s-electrons
- (d) They have one or more unpaired d-electrons**

Answer: (d) They have one or more unpaired d-electrons

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d0 and d10 species are diamagnetic.

124. Lanthanide contraction occurs because

- (a) f-orbital electrons are easily lost
- (b) f-orbital an incompletely filled
- (c) f-orbital electron an poor shielders of nuclear charge
- (d) f-orbital do not come out on the surface of atom and are buried inside**

Answer: (d) f-orbital do not come out on the surface of atom and are buried inside

Explanation: Across the lanthanoids, 4f electrons shield the nuclear charge poorly, so effective nuclear charge rises and atomic/ionic radii generally decrease.

125. Which is most reactive metal

- (a) Fe**
- (b) Pt
- (c) Ni
- (d) Co

Answer: (a) Fe

Explanation: Use the electrochemical series and the stability of the oxidation states involved to compare oxidising/reducing strength or reactivity.

126. Least reactive metal is

- (a) Fe
- (b) Os
- (c) Ni
- (d) Pt**



Answer: (d) Pt

Explanation: Use the electrochemical series and the stability of the oxidation states involved to compare oxidising/reducing strength or reactivity.

127. Which occludes hydrogen

- (a) Os
- (b) Pt
- (c) Ni
- (d) All of these**

Answer: (d) All of these

Explanation: The correct choice is All of these. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

128. Which has the maximum ferromagnetic character

- (a) Fe**
- (b) Co
- (c) Ni
- (d) Pt

Answer: (a) Fe

Explanation: The correct choice is Fe. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

129. Which forms interstitial compounds

- (a) Fe

(b) Co

(c) Ni

(d) All of these

Answer: (d) All of these

Explanation: The correct choice is All of these. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

130. Which occurs in nature in free state

- (a) Fe
- (b) Co
- (c) Ni
- (d) Pt**

Answer: (d) Pt

Explanation: The correct choice is Pt. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

131. $3d^{10}4s^0$ electronic configuration exhibits

- (a) Zn^{++}**
- (b) Cu^{++}
- (c) Cd^{++}
- (d) Hg^{++}

Answer: (a) Zn^{++}

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.



132. 3d04s0 electronic configuration exhibits

- (a) Pd+2
- (b) Sc+2
- (c) Ti+4**
- (d) Zn+2

Answer: (c) Ti+4

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.

133. Rare-earth elements are exhibited by

- (a) At. No. 58 to 71**
- (b) At. No. 21 to 30
- (c) At. No. 39 to 71
- (d) At. No. 81 to 91

Answer: (a) At. No. 58 to 71

Explanation: The correct choice is At. No. 58 to 71. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

134. All those elements belong to f - block whose atomic numbers are

- (a) and
- (b)
- (c) Both**
- (d) None

Answer: (c) Both

Explanation: The correct choice is Both. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

135. The correct order of density is

- (a) Cu>Ni>Zn>Sc
- (b) Ni>Cu>Zn>Sc**
- (c) Zn>Cu>Ni>Sc
- (d) Sc>Zn>Ni>Cu

Answer: (b) Ni>Cu>Zn>Sc

Explanation: Compare the relevant atomic masses and/or densities; the listed correct option has the greatest value among the choices.

136. The property exhibited by only transitional elements

- (a) To form paramagnetic compounds
- (b) To form coloured compounds
- (c) To form complex compounds**
- (d) To show inert tendency

Answer: (c) To form complex compounds

Explanation: The correct choice is To form complex compounds. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

137. Which of the following will have standard oxidation potential less than SHE



- (a) Zn
- (b) Cu
- (c) Fe
- (d) Ni

Answer: (b) Cu

Explanation: The correct choice is Cu. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

Explanation: The correct choice is Their 'd' orbitals are completely filled. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

138. Hydrated Cu^{2+} ion will be

- (a) Green
- (b) Violet
- (c) Blue
- (d) Colourless

Answer: (c) Blue

Explanation: The correct choice is Blue. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

140. Which of the following is the weakest reducing agent

- (a) Zn
- (b) Cu
- (c) H_2
- (d) Li

Answer: (b) Cu

Explanation: Use the electrochemical series and the stability of the oxidation states involved to compare oxidising/reducing strength or reactivity.

139. The placement of Zn, Cd and Hg along with 'd' block elements is not proper because

- (a) Their 'd' orbitals are completely filled
- (b) Their 'd' orbitals are empty
- (c) They do not form complex compounds
- (d) They do not form coloured compounds

Answer: (a) Their 'd' orbitals are completely filled

